

Statement concerning special BSc course project at DTU Environment

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Magnus Bøjlund Andersen, Mathias Werther, and Oliver Brincks Lund, all students at DTU Environment, made a special BSc course project 10 ECTS point in autumn 2021.

The project is named "Effects of UV radiation on plant growth and secondary metabolites." The purpose was to understand theoretically how UV light affects plant growth and assimilation of secondary metabolites. They work experimentally with a hydroponic growth system and LED lamps to deliver different spectral distributions. Plant effects were analyzed among other instruments with a new spectrometer tool, SpectraVue Leaf Spectrometer (CI-710), for absorption and reflection of indirect light measurement of secondary metabolites.

The students wanted to understand how to apply this knowledge to a vertical farming perspective to optimize the growth of plants using UV supplementation. They demonstrated that UV affected the pigment concentrations in plants.

The students delivered the mandatory report outlined as a scientific manuscript ready for publication. The quality is very high, and it fulfills all requirements for an "actual" scientific published paper, including graphically visualized data and information with proper statistical analysis. Only the low number of replicates, due to equipment restrictions, hinder a genuine submission to a scientific journal. They did high-class work and demonstrated a rare commitment to scientific work.

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